# Activity-4

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Given:-

- keys:- {4371, 1323, 6173, 4199, 4344, 9679, 1989}
- Hash function:- $h(x) = x \bmod 10$
- Hash table size:- 10, indexed 0-9

Initial hash values:-

| key | x mod 10 | Index |
|---|---|---|
| 4371 | 1 | 1 |
| 1323 | 3 | 3 |
| 6173 | 3 | 3 |
| 4199 | 9 | 9 |
| 4344 | 4 | 4 |
| 9679 | 9 | 9 |
| 1989 | 9 | 9 |

a. separate chaining.

| Index | hash table |
|---|---|
| 0 | - |
| 1 | 4371 |
| 2 | - |
| 3 | 1323 → 6173 |
| 4 | 4344 |
| 5 | - |
| 6 | - |
| 7 | - |
| 8 | - |
| 9 | 4199 → 9679 → 1989. |

b. open addressing - Linear Probing

| Index | Value |
|---|---|
| 0 | 9679 |
| 1 | 4371 |
| 2 | 1989 |
| 3 | 1323 |
| 4 | 6173 |
| 5 | 4344 |
| 6 | - |
| 7 | - |
| 8 | - |
| 9 | 4199 |

c. Open addressing - Quadratic probing

| Index | Value |
|---|---|
| 0 | 9679 |
| 1 | 4371 |
| 2 | - |
| 3 | 1323 |
| 4 | 6173 |
| 5 | 4344 |
| 6 | - |
| 7 | - |
| 8 | 1989 |
| 9 | 4199 |

d. Open addressing - Double hashing

| Index | Value |
|---|---|
| 0 | - |
| 1 | 4371 |
| 2 | - |
| 3 | 1323 |
| 4 | 4344 |
| 5 | 9679 |
| 6 | - |
| 7 | 6173 |
| 8 | - |
| 9 | 4199 |